



Hyperloop Interior Design Guidelines

Final Report

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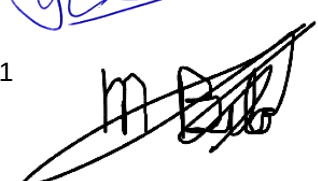
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Summary

Hardt is developing Hyperloop, a new transportation system for passengers and cargo. Pressurized vehicles travel inside a low-pressure tube by means of a magnetic levitation and propulsion system. Hardt's goal is to create a European hyperloop network, capable of connecting the 50 largest cities in Europe and enable high-speed sustainable travel.

For the passenger version of Hyperloop, Hyperloop Development Program (HDP) has developed a concept interior. ADSE was requested to use its 25 years of experience in both rail and aviation to evaluate this interior concept against aerospace 'best practices'. This evaluation was performed under a 3-year support agreement between Hyperloop and ADSE.

With this report, ADSE delivers a "checklist" of the most relevant interior topics that can be used by HDP as interior design guidelines. In addition, ADSE points out that several important decisions are still to be made by HDP. Examples are the decision to integrate lavatories or not, and the concept for emergency evacuation. Both have a large impact on the Hyperloop interior and therefore need to receive adequate attention early in the project.

1 Introduction and Purpose

ADSE was asked by Hyperloop Development Program (HPD) to perform an evaluation of the interior concept that has been developed by HPD for a passenger version of Hyperloop. For this review, ADSE was requested to combine its rail experience with ‘best practices’ from aerospace. These best practices can be split into a *physical* perspective (e.g. what is the best way to attach the seats to the structure?) and an *organizational* perspective (e.g. how do we include certification aspects in the design from the very start of the project?)

In preparation for this evaluation, ADSE has shared an Intermediate Report with HDP in May 2021 (21-RA-012). The purpose of this report was to check that the information we collected was in accordance with the latest insights. This information is used as input for this final report.

The goal of the project was to evaluate the maturity of the concept design and to identify topics that still need more attention. After several internal discussions, ADSE decided that it would be most valuable to deliver an overview of most relevant interior topics that can be used as a checklist by HDP. As a result, this document contains the **Interior design guidelines for HDP**.

With that, this final report has the following purpose:

- To present ADSE’s approach to come to the interior design guidelines.
- To share an extensive checklist of relevant interior topics and guidelines for HDP to include in the design of the interior of the Hyperloop.
- To point-out any points of concern that ADSE has at this stage of the design.

Note: In this document, any reference to Hyperloop refers to the *passenger* version of the vehicle.

2 Approach

2.1 Input Documents

For ADSE to perform this evaluation, HPD has shared an extensive list of documents related to the Hyperloop interior. This list is given in Annex 1.

After a detailed scan, ADSE has used the following documents as main input for this report:

No.	Title	Date	Topic
08	Acceptance of Hyperloop	Nov 2020	Hyperloop acceptance among potential users
15	Hyperloop Interior	-	Requirements & concept design
16	Hyperloop parameters [Excel document]	-	Operational Control Concept
22	Roland Berger for Kingdom Saudi Arabia	24-03-21	Broad system overview
23	Service Design Sprint Report	-	Stakeholder Needs
24	The Hyperloop	13-09-19	Hyperloop Vision document

2.2 Steps Taken

For this interior evaluation, ADSE has taken the following steps:

1. ADSE has studied all documents shared by HPD for input that is related to the Hyperloop interior,
2. A summary was made of the interior concept and shared with HDP in an Intermediate Report (ADSE document 21-RA-012) to check that the collected information was in accordance with the latest insights,
3. The stakeholders of the entire lifecycle of the Hyperloop interior were identified,
4. The needs and expectations of each stakeholder were defined,
5. Based on the stakeholder analysis, ADSE has created a 'checklist' of all the topics that need to be addressed in the design of the Hyperloop interior,
6. Topics of concern were identified and summarized in conclusions and recommendations.

The next chapters describe the results of these steps.

3 Stakeholder Analysis

The interior design of the Hyperloop should satisfy the needs and expectations of all relevant stakeholders. ADSE has used a 'systems engineering approach' to make sure the list of stakeholders is complete and all relevant topics are addressed.

This chapter identifies these stakeholders based on a typical lifecycle of the Hyperloop interior: from concept design to the end of its service life. Subsequently, the needs and expectations are defined for each of them. This stakeholder analysis is the starting point to define the functional requirements for the Hyperloop interior.

3.1 Lifecycle of Hyperloop Interior

In general, the lifecycle of a product starts with its design, via manufacture and use (operation and maintenance), until the product's removal from service and disposal (end of life)¹. This is shown schematically in Figure 1.

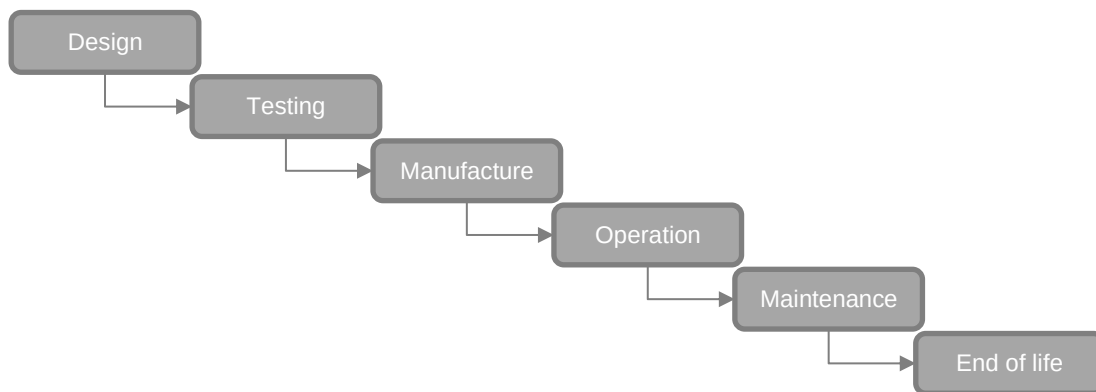


Figure 1: Life cycle of a product

Even though most of these stages occur more or less sequentially over the life cycle of a product, the design must take all of these phases and their interrelationship into account.

In the lifecycle of the Hyperloop, the following key stakeholders are identified sequentially:

1. HDP board and investors
2. Interior designer
3. Engineer
4. Certification staff
5. Production department
6. System suppliers
7. Passenger
8. Cleaning crew
9. Maintenance personnel
10. Recycling company

¹ Michael Grieves, Product Lifecycle Management: Driving the Next Generation of Lean Thinking, 2006

The ecosystem of the Hyperloop interior is defined based on so-called 'Operational Entities', components of the Hyperloop, and stakeholders, as depicted in Figure 2. The items on each level share interfaces and can be an element of multiple entities on lower levels; e.g. the Propulsion system (Level 2) is a component of the Hyperloop pod, but also an aspect of the Hyperloop track (Level 1).

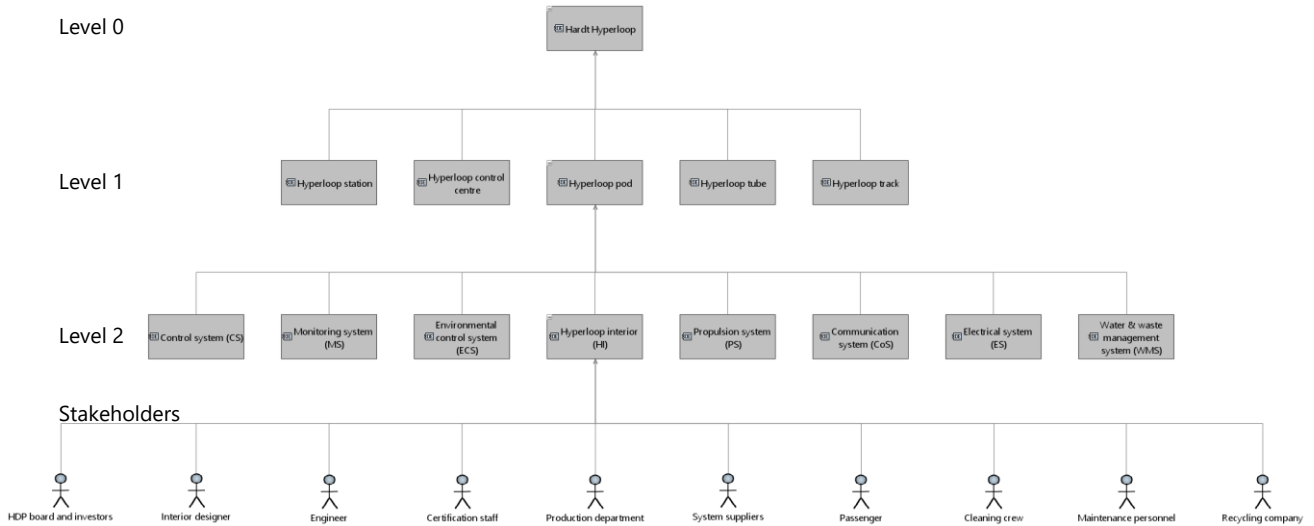


Figure 2: Ecosystem of the Hyperloop interior

In the next paragraph, the needs and expectations of each stakeholder are addressed.

3.2 Needs and Expectations

Each stakeholder in the lifecycle of the Hyperloop has its own role for the realization of the final product, as summarized in Table 1.

Table 1: Overview of the stakeholders and their role

Number	Stakeholder	Role
1.	HDP board and investors	Provide the correct circumstances and input to create and use the Hyperloop interior.
2.	Interior designer	Define the interior lay-out, aesthetics and the travel experience offered to the passenger of the Hyperloop.
3.	Engineer	Translate the concept of the Interior designer into a technical design that can be produced, certified, used and maintained.
4.	Certification staff	Certify the Hyperloop interior.
5.	Production department	Produce and assemble the Hyperloop interior.
6.	System suppliers	Develop and supply (sub)systems for the Hyperloop interior.
7.	Passenger	User the Hyperloop interior.
8.	Cleaning crew	Clean the Hyperloop interior.
9.	Maintenance personnel	Maintain the Hyperloop interior.
10.	Recycling company	Disassemble and recycle/dispose the Hyperloop interior.

The needs and expectations of each stakeholder are described in corresponding subsections. These in-depth analyses are substantiated using a so-called ‘Operational Capabilities Breakdown’, which illustrates the functions that a certain stakeholder expects from the Hyperloop interior (the Operational Entity). These functions are derived by considering relevant scenarios for each stakeholder, like the typical journey of a passenger. Furthermore, such diagrams provide a clear overview of the various interactions, relationships and interfaces between the needs of the stakeholders, Operational Entities and Operational Capabilities.

The capabilities identified in these subsections serve as input for the remainder of this interior evaluation, as stated in Paragraph 2.2.

3.2.1 HDP board and investors

Figure 3 contains a breakdown of Operational Capabilities (OC's) for the HDP board and investors. It can be observed that the several expectations of the HDP board and investors coincide with the capabilities of the interior expected by the Production department (5), System suppliers (6), and the Passenger (7).

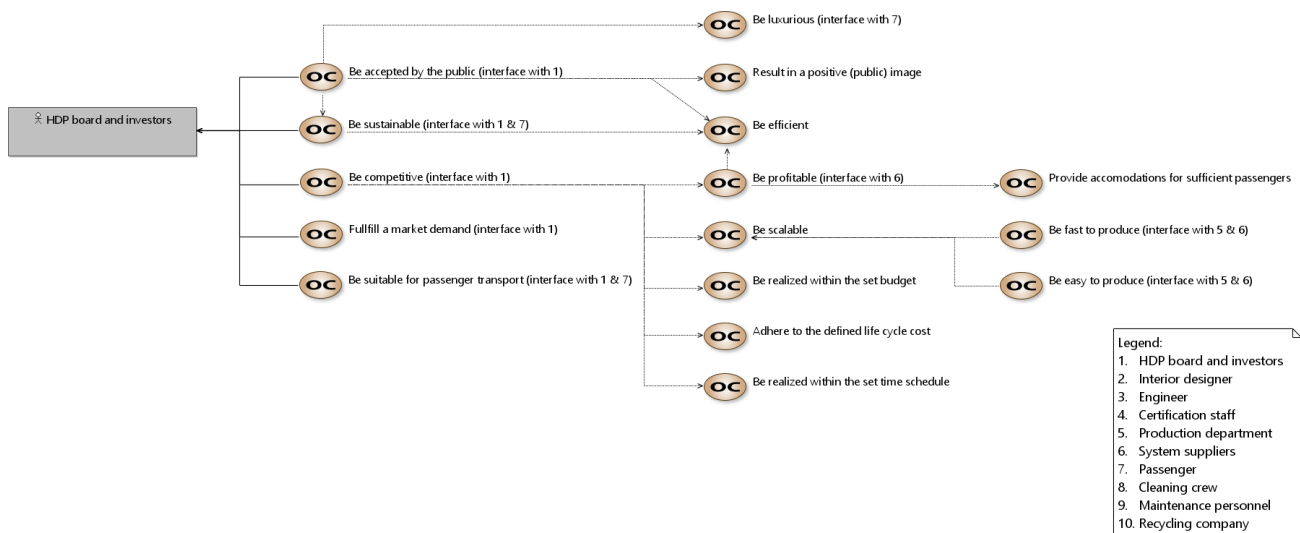


Figure 3: Operational Capabilities Breakdown for the HDP board and investors

3.2.2 Interior designer

In this report, the Interior Designer focusses on the interior lay-out, aesthetics and the travel experience offered to the passenger. More technical topics like climate control, lighting, anthropometrics, ergonomics and emergency procedures are considered to be part of the scope of the Engineer.

The Operational Capabilities Breakdown for the Interior designer is depicted in Figure 4. The Hyperloop interior should provide aesthetic freedom for this stakeholder, which conflicts with a multitude of needs from other stakeholders. For clarity, only one of such conflicts is indicated: the freedom in materials. The desired capability of aesthetic freedom and the resulting conflicts should consistently and properly be taken into account during the Hyperloop interior design.

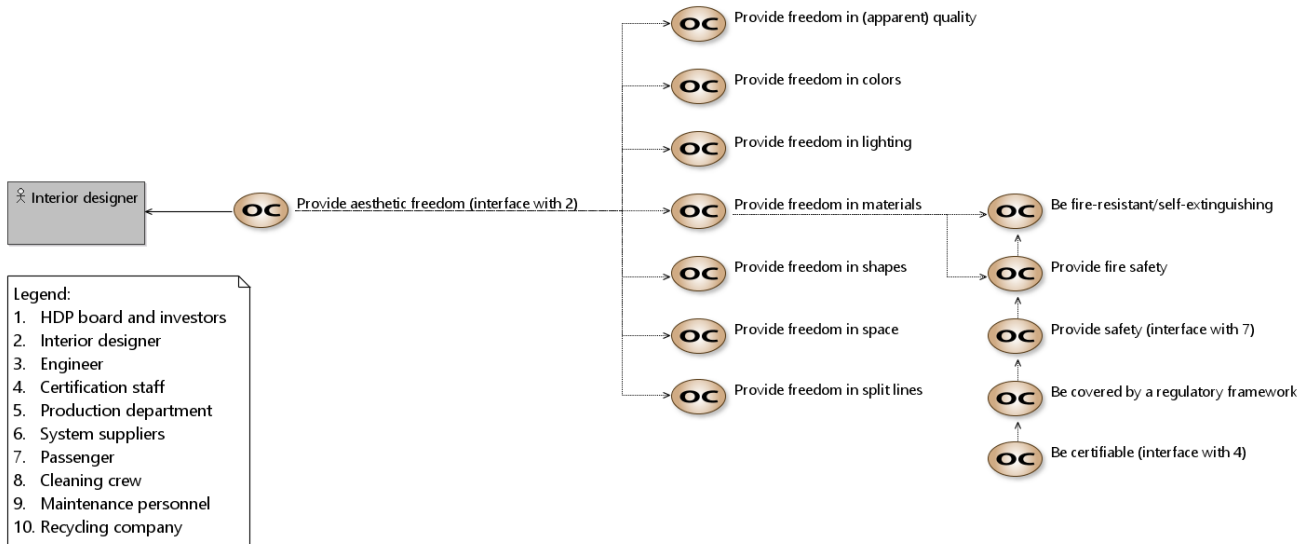


Figure 4: Operational Capabilities Breakdown for the Interior designer

3.2.3 Engineer

Figure 5 illustrates the Operational Capabilities Breakdown for the Engineer designing the Hyperloop interior. The engineer translates the concept of the Interior designer into a technical design that can be produced, certified, used and maintained.

Several of the capabilities coincide with the needs identified for the HDP board and investors (1), Certification staff (4) and Passenger (7). Although the requirements for the Hyperloop interior taken into account by the Engineer are driven by the needs of other stakeholders as well, the capabilities of the interior from which these requirements follow are not relevant for the Engineer of the Hyperloop interior and therefore not indicated as relationship

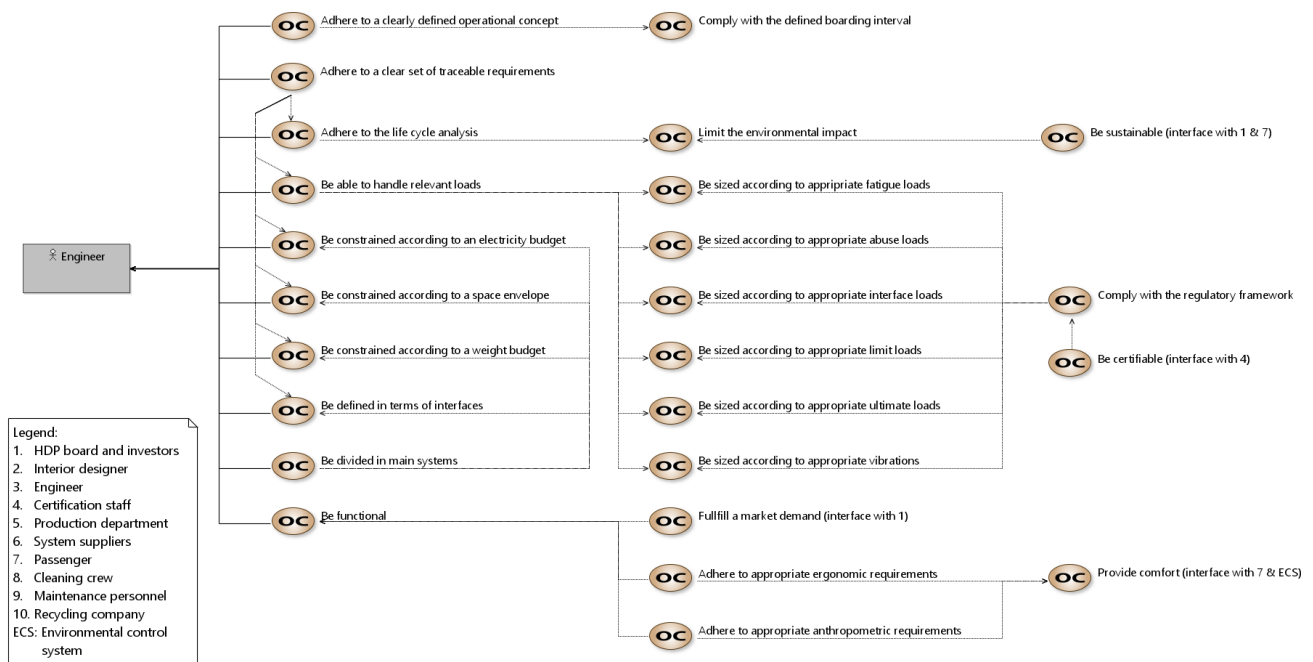


Figure 5: Operational Capabilities Breakdown for the Engineer

3.2.4 Certification staff

For the Certification staff, the Operational Capabilities Breakdown is shown in Figure 6. The functions of the interior required by this stakeholder contain a multitude of capabilities aligned with needs of other stakeholders (Production Department (5), System suppliers (6), and Passenger (7)) and several interfaces with Operational Entities (Communication system, Propulsion system and Electrical system) of level 2 (Figure 2).

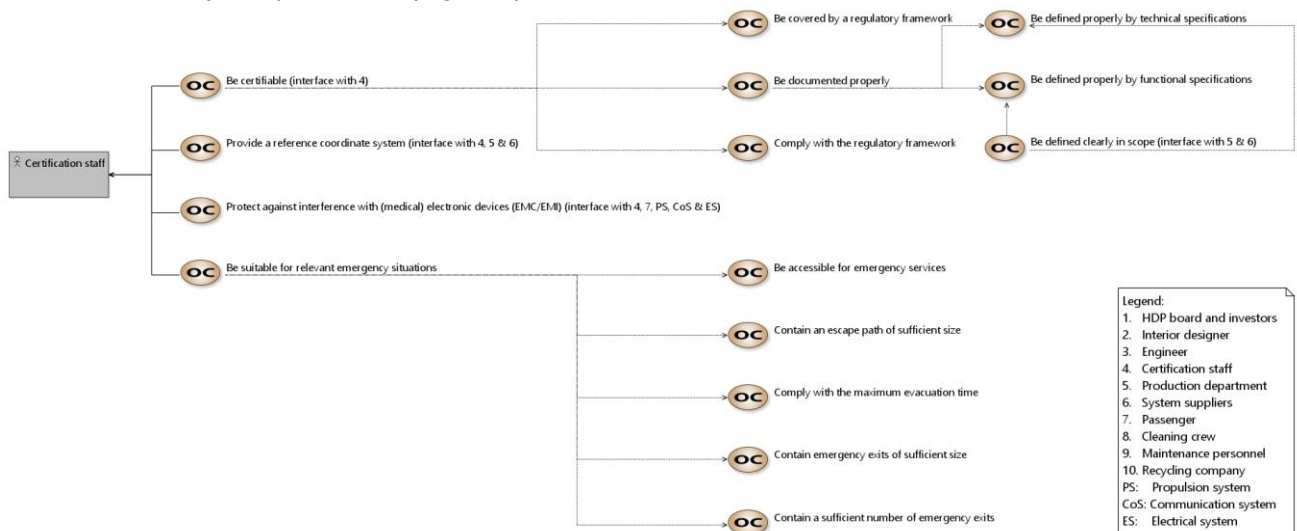


Figure 6: Operational Capabilities Breakdown for the Certification staff

3.2.5 Production department

Figure 7 shows the Operational Capabilities Breakdown for the Production department. Even though the number of capabilities of the interior required by the Production department is limited, each one is aligned with capabilities expected by other stakeholders (Certification staff (4), System suppliers (6), Maintenance personnel (9), and Recycling company (10)).

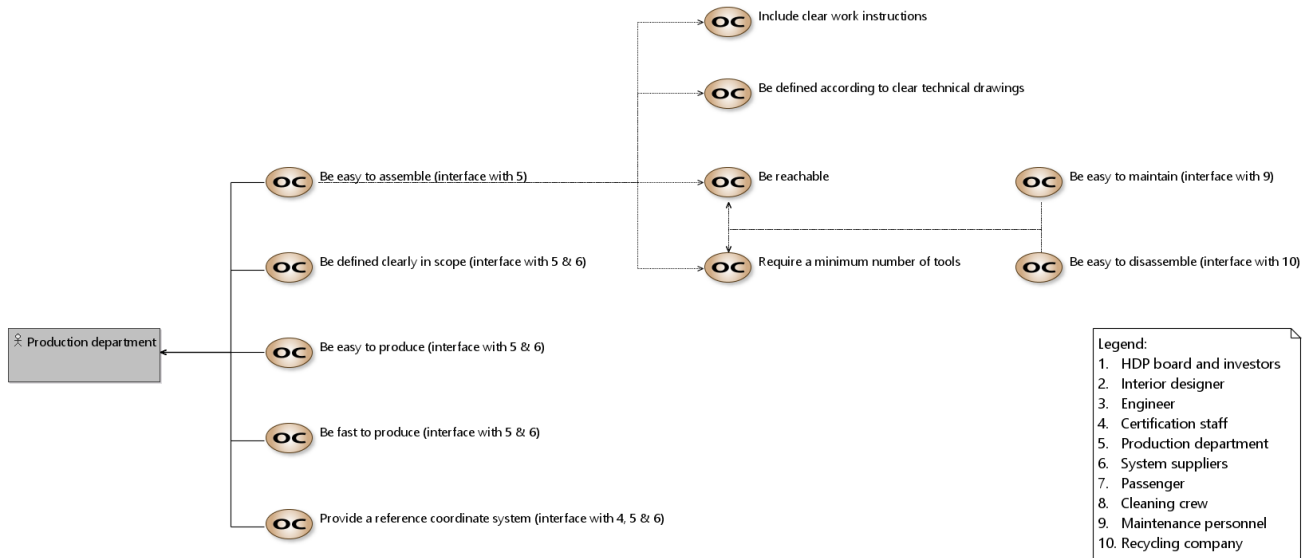


Figure 7: Operational Capabilities Breakdown for the Production department

3.2.6 System suppliers

The Operational Capabilities Breakdown for the System suppliers is illustrated in Figure 8. It includes several functions of the interior which are aligned with the expectations of the HDP board and investors (1), Certification Staff (4), and Production department (5).

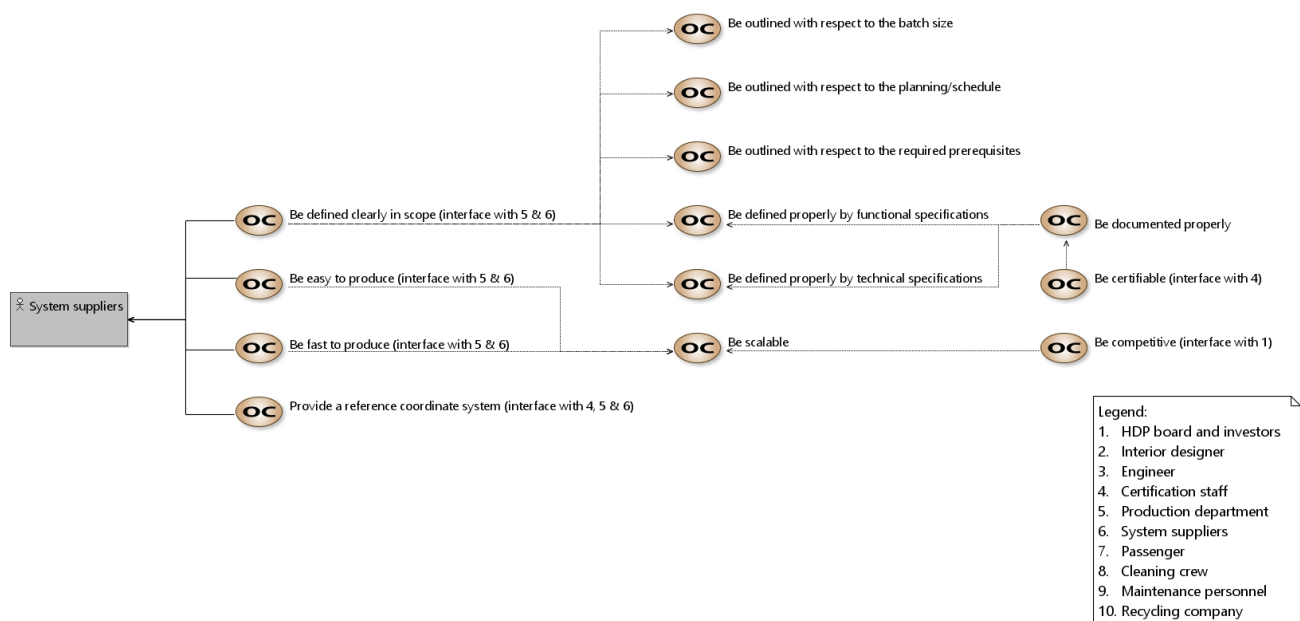


Figure 8: Operational Capabilities Breakdown for the System suppliers

3.2.7 Passenger

In Figure 9, the Operational Capabilities Breakdown for the Passenger can be seen.

Several of the capabilities coincide with the needs identified for the HDP board and investors (1), Interior designer (2), Certification staff (4), and the Cleaning crew (8). Lastly, several interfaces with each of the other components of the Hyperloop pod (Level 2 in Figure 2) can be identified. This, in combination with the number of capabilities expected from the Hyperloop interior by the Passenger, is as expected since the Passenger is one of the primary stakeholders of the interior.

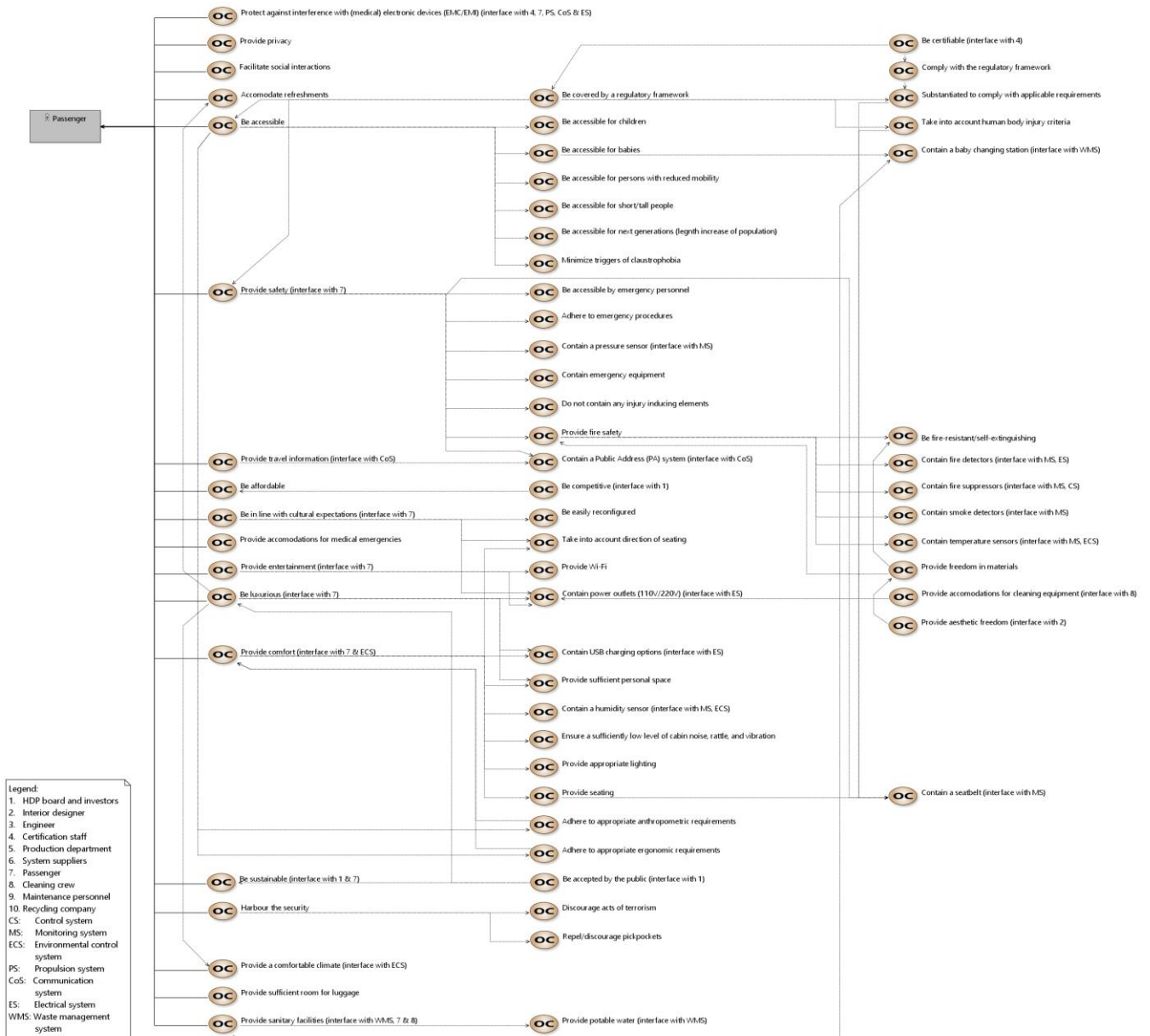


Figure 9: Operational Capabilities Breakdown for the Passenger

3.2.8 Cleaning crew

For the Cleaning crew, the Operational Capabilities Breakdown is detailed in Figure 6. Interfaces with the Electrical system and the Water & waste management system (Figure 2) are shown in this figure. Lastly, several capabilities are in agreement with the expectations of other stakeholders (HDP board and investors (1), Passenger (7), and Maintenance personnel (9)) for the Hyperloop interior.

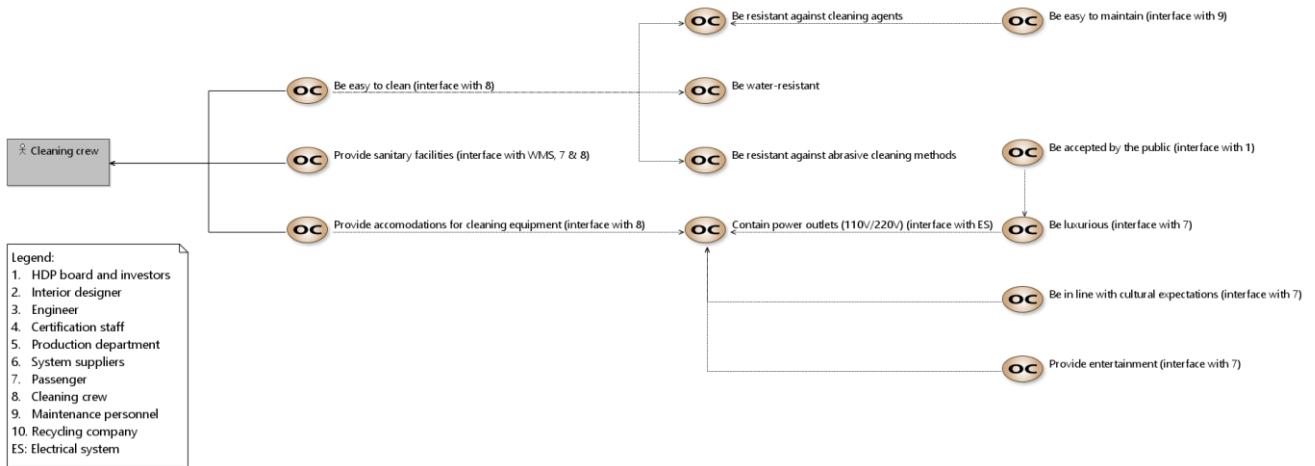


Figure 10: Operational Capabilities Breakdown for the Cleaning crew

3.2.9 Maintenance personnel

The Operational Capabilities Breakdown is displayed in Figure 11 for the Maintenance personnel. It depicts an interface with the Monitoring system (Figure 2) and several coinciding capabilities with the Production department (5), Cleaning crew (8), and Recycling company (10).

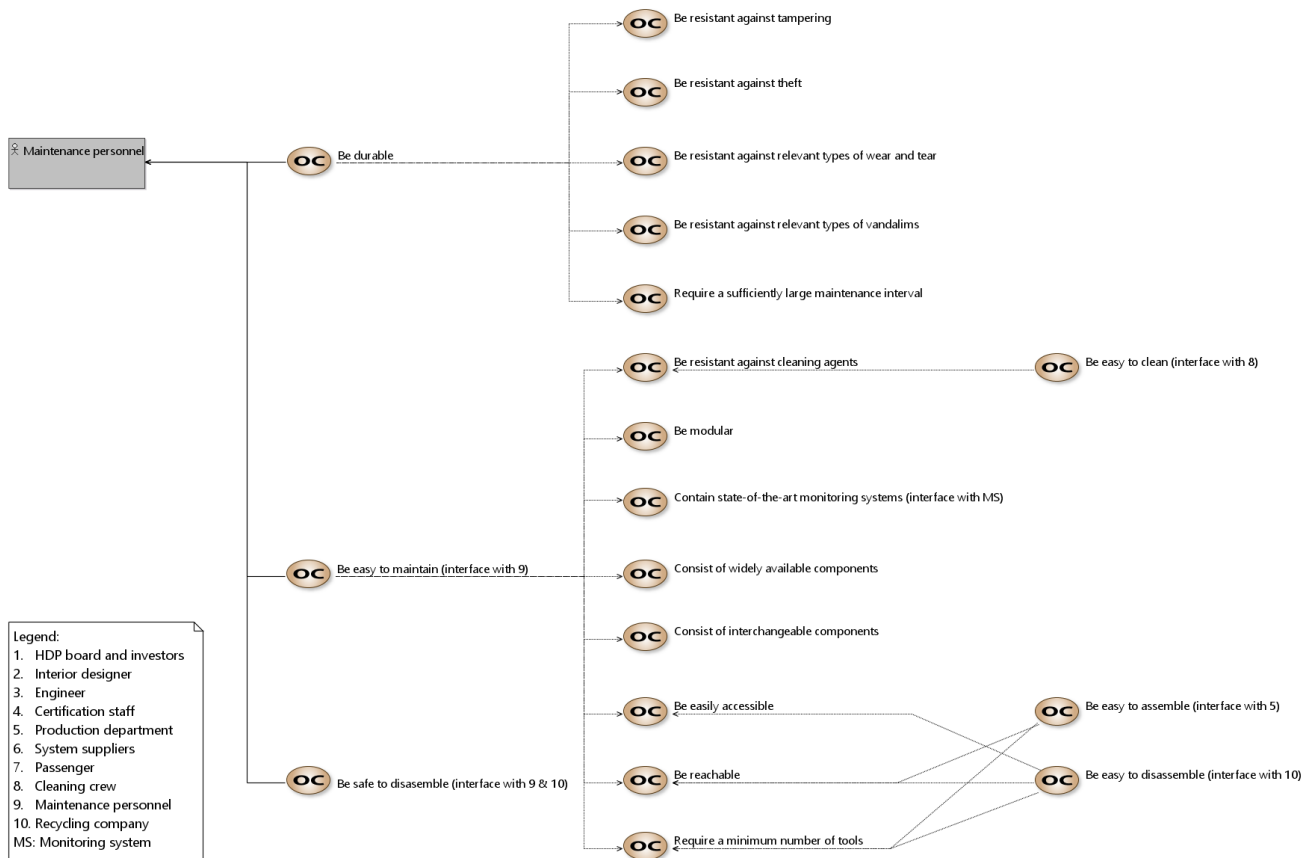


Figure 11: Operational Capabilities Breakdown for the Maintenance personnel

3.2.10 Recycling company

Figure 12 contains the Operational Capabilities Breakdown for the Recycling company. The diagram contains several capabilities of the interior which are also expected by the Production department (5) and Maintenance personnel (9).

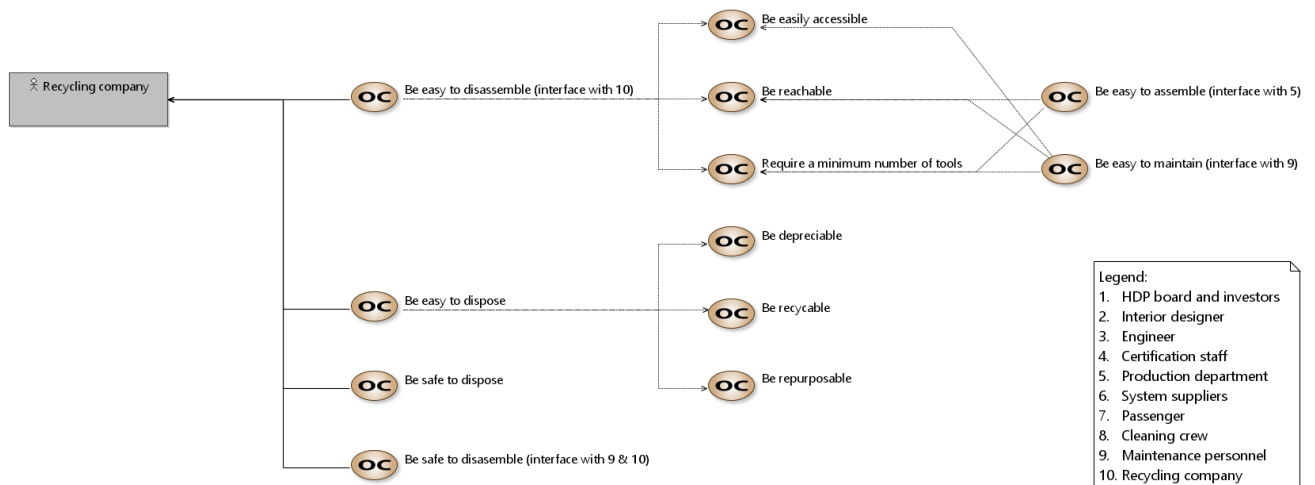


Figure 12: Operational Capabilities Breakdown for the Recycling company

4 Interior Requirements Checklist

As mentioned in the Introduction of this report, ADSE has created an overview of the most relevant interior topics that can be used as a checklist by HDP. This checklist is a practical derivative of the stakeholder analysis in Chapter 3. This chapter gives an overview of these topics.

The checklist is split into eight different categories:

1. Top-Level
2. Interior
3. Operational
4. Structural
5. Electrical Systems
6. Environmental Control System (ECS)
7. Certification & Safety
8. Miscellaneous

Each requirement or point of attention has a reference number and a priority. These priorities have the following meaning:

Priority 1: These topics strongly influence the entire design and need to be frozen early in the process. Changes to these topics at a later stage will have a significant impact on cost and planning of the entire project.

Priority 2: These topics must be defined at the end of the concept design phase, or early during the detail design. Since they typically have an interface with other parts or systems, or impact on certification, changes may have a wide effect on the overall design.

Priority 3: These topics have limited or no interface to other parts or systems, or are related to the operation of the Hyperloop. They can be defined at a late stage in the design process.

The letters CL in the numbering of each item stand for 'Checklist'.

4.1 Top Level

No.	Prio.	Description
CL.1.1	1	Against what standards shall the design being created? What needs to be included in the certification plan?
CL.1.2	1	What passenger functions are integrated? For instance boarding, seating, multiple classes, luggage storage, lavatories, catering?
CL.1.3	1	What is the desired payload weight vs empty weight ratio? This influences the length for an emergency brake, and puts restrictions on the weight of the interior.
CL.1.4	1	What is the cabin length and diameter?
CL.1.5	1	What is the number of passengers and seats per vehicle?
CL.1.6	1	What will be the lay-out of the passenger accommodation (LOPA): doors, seats, aisles, monuments, luggage, lavatory, evacuation route.
CL.1.7	1	What systems need to be integrated and what is their space envelope and weight budget? For instance ECS, ducting and electrical systems.
CL.1.8	1	What is the coordinate system for references and axes (x,y,z) in order to define and communicate space allocations and interfaces? Where is the 'zero' point?

4.2 Interior

No.	Prio.	Description
General Cabin Lay-Out Aspects		
CL.2.1	1	What will be the location of exits, aisles, seats and partition walls?
CL.2.2	1	Must the cabin lay-out be easily reconfigurable (seat pitch, partition locations)? This influences the floor structure and seat attachments (seat tracks?)
CL.2.3	1	Will there be sidewall tables installed in club-seating configuration (seats facing each other)? How will injury be prevented in case of abrupt deceleration?
CL.2.4	1	What will be the location and size of luggage bins and stowages? What luggage type, size and weight shall they be able to hold?
CL.2.5	1	Will there be internal doors? If yes: - In case of swing doors: How will they be kept in a neutral position when not in use? Will they be able to keep their neutral position during emergency brake, or will the doors swing open and create a hazard to occupants? - In case of sliding doors: Can they introduce a risk during emergency evacuation? Can they get jammed? What about rapid decompression?
CL.2.6	1	Will there be (integrated) waste bins in the interior? Will they be in the in (or near) every seat, or centralized waste bins? Is waste separation a topic of concern?
CL.2.7	1	Is there a separate area for train crew foreseen?
CL.2.8	1	Are baby-care facilities foreseen? If yes, then how many and where?
CL.2.9	1	How will persons with reduced mobility (PRM) be accommodated?
CL.2.10	1	How to minimize the feeling of claustrophobia?
Seating		
CL.2.11	1	Direction of seats: forward and backward facing? Also sideward facing?
CL.2.12	1	Can seats be turned around at terminal stations? (Asians like to travel forward)
CL.2.13	1	Will there be tip-up seats installed in locations where a flexible interior is desired? E.g. near a wheel chair location?
CL.2.14	1	What will be the seat measures, depending on anthropometric data of human body. Notes: - Take into account the increasing length of the population - Take into account children (>2 years old)
CL.2.15	1	Differences in size and seat features between first and second class seats?
CL.2.16	1	Under-seat luggage stowage?
CL.2.17	1	Seat recline?
CL.2.18	2	Arm rests (movable?)
CL.2.19	2	Head rest (movable?)
CL.2.20	2	Foot rest?
CL.2.21	2	Tray table or in-arm table? With cup holder?
CL.2.22	2	Phone or tablet holder options?

CL.2.23	2	Reading lights?
CL.2.24	2	Literature pockets?
CL.2.25	2	On-board entertainment (screen in seat)?
CL.2.26	2	USB charging options and power outlet (220V)?
CL.2.27	2	Coat hook integrated in seat? Or in the interior?
CL.2.28	2	Will there be 'fasten seatbelt signs' in the seats? Or in the interior?
CL.2.29	2	Is a footstep on the seats for a short person to look into luggage bins necessary?
Aesthetics		
CL.2.30	2	What will be the surface texture and finish of the interior lining (sidewall panels and ceiling), partitions, monuments and seat upholstery?
CL.2.31	2	Can the aesthetic aspects (the 'look and feel') of the interior be easily adjusted to the requirements of different customers (color schemes, logo's)? I.e. Branding?
CL.2.32	2	What is the minimum width of split lines between panels to account for movement on the interior during operation?
CL.2.33	3	Will there be expressions of art on sidewalls, partitions, glass panels or doors?

4.3 Structure Aspects

No.	Prio.	Description
CL.3.1	1	Against which loads shall the interior, the seats and the attachments be designed? What are the interface loads, abuse loads, limit loads, ultimate loads? Take into account survivable emergency conditions (human body injury criteria).
CL.3.2	1	Definition of structure for floor, sidewalls and ceilings.
CL.3.3	1	Installation concept of interior: Back-up structure, tie rods, brackets?
CL.3.4	1	Installation concept of luggage bins: Accessibility of attachment points, interface with electrical system (reading lights) and/or ECS (air outlets)?
CL.3.5	1	Installation concept of ECS ducting: What space envelope is defined and which attachment points need to be available?
CL.3.6	1	Interface between ECS ducts and interior: How is air distributed within the cabin?
CL.3.7	1	How are the monuments (e.g. lavatories and stowages) structurally attached to floor and ceiling?
CL.3.8	1	Installation concept of seats: Wall-mounted or floor mounted? Seat track or fixed location?
CL.3.9	2	Permanent deformation of the seat in case of a catastrophic event shall stay within limits to keep an adequate emergency escape path.
CL.3.10	2	Are firm hand holds along the aisle necessary to move around the cabin? How are they integrated in the interior? Part of the seats?
CL.3.11	2	How is the interior designed to withstand vibration loads and fatigue?
CL.3.12	2	Is it necessary to attach the lining panels to the structure via vibration isolators to reduce cabin noise?

4.4 Electrical Systems

No.	Prio.	Description
CL.4.1	1	Which electrical systems share an interface and possibly have to communicate with each other?
CL.4.2	1	What is the electrical power budget for each electrical system?
CL.4.3	1	What power converters are necessary to supply power to the various electrical systems? Like the systems below.
CL.4.4	1	Environmental Control System and Pressurization of cabin
CL.4.5	1	Cabin lighting: Does its color adapt to the time of day or other factors?
CL.4.6	1	Illuminated signs
CL.4.7	1	Emergency escape path marking
CL.4.8	2	Power outlets (220V and/or USB)
CL.4.9	2	Wi-Fi
CL.4.10	2	Passenger Address (PA) system and speakers
CL.4.11	2	Seatbelt monitoring system
CL.4.12	2	Temperature sensors
CL.4.13	2	Smoke detectors
CL.4.14	2	Surveillance cameras
CL.4.15	2	On-board entertainment and passenger information systems
CL.4.16	1	If a lavatory is installed: Water tank, pump, vacuum system, bio-reactor

4.5 Environmental Control System (ECS)

No.	Prio.	Description
Technical and Performance		
CL.5.1	1	What is the required performance of the ECS system to meet requirements for pressurization, heating, cooling and fresh air?
CL.5.2	1	What redundancy is necessary for the ECS system in case of failure, since the pressurization of the pod depends on this system?
CL.5.3	2	What are acceptable flow rates in ducts, in air outlets and in the cabin to keep noise and draft to a minimum? This influences duct diameters and the design of air outflow plenums. See also CL.8.8.
CL.5.4	2	What are acceptable noise levels of fans? A constant noise is typically better than an intermitting noise. See also CL.8.8.
CL.5.5	2	Are individual gasper air outlets installed for the passenger to be in control of fresh air?
CL.5.6	2	Are HEPA filters installed to filter recirculated air?

Comfort Aspects		
CL.5.7	2	Constant temperature throughout the cabin.
CL.5.8	2	Relative humidity at comfortable level? Note: How to prevent condensation on cold surfaces and air conditioning ducts?
CL.5.9	2	How are changes in pressurization level are kept to a minimum?
CL.5.10	2	Noise produced by the ECS system, and the air through the ducts and outflow plenums is kept as low as possible.

4.6 Certification and Cabin Safety

No.	Prio.	Description
General		
CL.6.1	1	Against which standards shall the design being created, and what will be the certification plan?
Emergency Scenarios		
CL.6.2	1	Which emergency scenarios can occur? Is there also a collision scenario?
CL.6.3	1	How is rapid decompression in case of a big air leak incorporated in the design?
CL.6.4	1	Which electrical back-up systems are present in case of power loss?
CL.6.5	2	Emergency lighting and escape path marking shall automatically switch on in case of loss of electric power.
CL.6.6	1	Is the number of emergency exits, width of escape path and size of emergency exits sufficient for maximum evacuation time? In aviation, this is 90 seconds.
CL.6.7	1	Which emergency exits can be used during which emergency scenario?
CL.6.8	1	Possible "dead ends" in the cabin are dangerous during evacuation. Passengers can potentially get trapped on their way to evacuate, especially with smoke in the cabin or limited visibility.
CL.6.9	3	Are exit location signs visible from main aisle?
CL.6.10	3	Instructions for exit opening must be indicated on, or near, emergency exits.
CL.6.11	3	Emergency exits must have exterior marking for emergency personnel.
CL.6.12	2	Emergency oxygen: Are emergency oxygen masks necessary above each seating position in case of loss of pressurization? And <i>portable</i> oxygen, so passengers can move freely, for instance in case of fire? For how many minutes does the emergency oxygen need to last? Note that the oxygen generators get very hot during use.
CL.6.13	1	Emergency evacuation: How to deal with low-pressure environment of the vacuum tube?
CL.6.14	1	Equipment/cargo/large masses may not cause injury or hinder escape facilities during an emergency evacuation.
CL.6.15	2	Is there a possible obstruction caused by arm rests or tray tables during emergency evacuation?

Fire Safety		
CL.6.16	2	Materials, parts, equipment and wiring must comply to fire safety standards (e.g. self-extinguishing).
CL.6.17	2	Smoke detection must be installed in closed areas like lavatories.
CL.6.18	2	Automatic fire suppression must be installed in waste compartments.
CL.6.19	2	Fire detection and suppression systems must be installed in cargo area if not accessible by passengers/crew during trip.
CL.6.20	2	Cargo area must have fire self-containment/burn through resistance.
Means of Compliance		
CL.6.21	2	How will the requirements for certification and cabin safety be demonstrated (Means of Compliance)?
CL.6.22	2	Is dynamic testing of seats and seat attachments necessary?
CL.6.23	3	What test labs, test rigs and set-ups will be required for compliance demonstration?

4.7 Operational

No.	Prio.	Description
CL.7.1	2	What is the operational concept? Boarding and deboarding time, passenger logistics, check-in luggage, carry-on luggage.
CL.7.2	2	What indications (visible and audible) are foreseen to signal to passengers that the doors are closing?
CL.7.3	2	Will there be an emergency brake that can be operated by passengers? And how will the communication between passengers and the outside world take place if there is now crew on board?
CL.7.4	2	On-board beverage or snack service? Or on-board vending machines?

4.8 Miscellaneous Topics

No.	Prio.	Description
CL.8.1	1	How is 'Design for Manufacturing and Assembly' incorporated in the design? I.e. optimization of the design to create it cheaper and more efficiently.
CL.8.2	1	How is 'Design for Maintenance' incorporated in the design? E.g. modular, easily accessible, minimum tools necessary, long inspection intervals, high durability.
CL.8.3	1	If a lavatory is installed: Will it include a urinal?
CL.8.4	1	If a lavatory is installed: What is the desired service interval and what will be the service procedure?
CL.8.5	2	What to expect with respect to wear and tear, and vandalism?

CL.8.6	2	How is Sustainability integrated in the design? I.e. use of sustainable materials and recycling.
CL.8.7	1	How is 'Design for Cleanability' incorporated in the design. E.g. Free floor without seat supports, water resistant, round corners and no pockets where dirt can accumulate. Also resistance against common beverages like coffee, tea and soft drinks, and resistance against chemical cleaning agents shall be considered.
CL.8.8	2	How is cabin noise kept to a minimum? E.g. rattle and vibration, noise of interior systems, the ECS airflow and noise penetrating from outside. Is there enough acoustic damping inside the cabin to absorb passenger noise? Are sidewall and/or ceiling panels treated or perforated to improve acoustic absorption of noise.
CL.8.9	2	How is thermal insulation for a constant climate in the cabin integrated in the design?
CL.8.10	3	What lighting levels and color temperature must be chosen, also in relation to passenger comfort?
CL.8.11	2	Is the floor covering slip-resistant? And is dirt masked by smart choice of colors and patterns?
CL.8.12	2	Is the topic of finger pinching (entrapment) between doors, recline mechanisms, tray tables and arm rests sufficiently addressed?
CL.8.13	2	Is the possibility of injury by protruding structure or sharp edges avoided as much as possible?
CL.8.14	2	Are luggage bins designed such that luggage can't fall out, also during emergency brake?
CL.8.15	2	Can lavatory doors (if applicable) be unlocked from the outside? And, if they open into the aisle, do they close in evacuation-flow direction to prevent obstruction during emergency evacuation?
CL.8.16	2	Does the interior facilitate the search for suspicious objects (terrorism). I.e. no possibilities to hide objects in the cabin.
CL.8.17	3	How are seat assignment placards or illuminated signs integrated in the design?
CL.8.18	3	Are fasten seat belt signs visible from all seats?

5 Cabin Safety and Certification Aspects

This chapter addresses some important cabin safety and certification aspects of the Hyperloop interior in more detail and are an addition to the checklist in Chapter 4. Since no regulatory framework is currently in place, the interior is evaluated against existing aerospace and rail regulations.

5.1 Fire Safety, Smoke Detection and Fire Suppression

For airplanes, the following materials, parts and areas need to comply to flammability requirements:

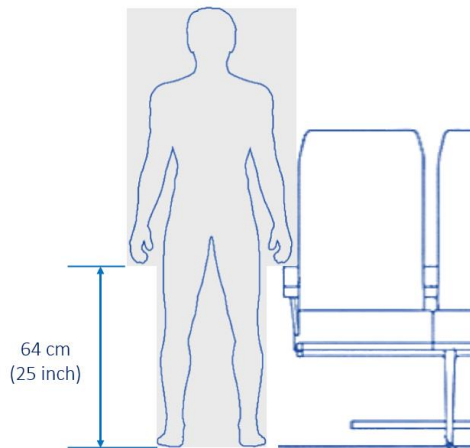
1. All interior panels like ceilings, sidewall panels, partitions and cabinet walls
2. Stowage compartments, furnishings, shelves, containers and bins
3. Transparencies (like illuminated signs), molded & thermoformed parts
4. Seat cushions, padding, fabrics & leather
5. Seat belts and straps
6. Structural flooring and floor covering
7. Electrical conduit, wiring & components
8. Air ducting, air ducting joints and trim strips
9. Thermal and acoustic insulation materials
10. Cargo compartment linings

5.2 Emergency Evacuation

Access to the emergency exits:

- Emergency escape path marking shall guide passengers to the exits when smoke has filled the cabin.
- Exit signs shall clearly mark the location of each exit.
- The cabin shall have no “dead ends” where passengers can get trapped in case of low visibility conditions.
- Operation of doors by passengers shall be clearly marked.

For airplanes, the passenger aisle width at any point between seats must equal to or exceed the values in the following table:



Passenger seating capacity	Minimum passenger aisle width (cm (inches))	
	Less than 64 cm (25 inches) from floor	64 cm (25 inches) and more from floor
10 or less	30 (12)*	38 (15)
11 to 19	30 (12)	51 (20)
20 or more	38 (15)	51 (20)

5.3 Emergency Equipment

For aerospace, the typical set of emergency equipment carried on board is shown in Figure 1. Since Hyperloop passengers are traveling in a pressurized cabin, just like in an aircraft, most of this equipment is applicable and must be included in the cabin design. Obviously, life vests and rafts are not applicable for Hyperloop.

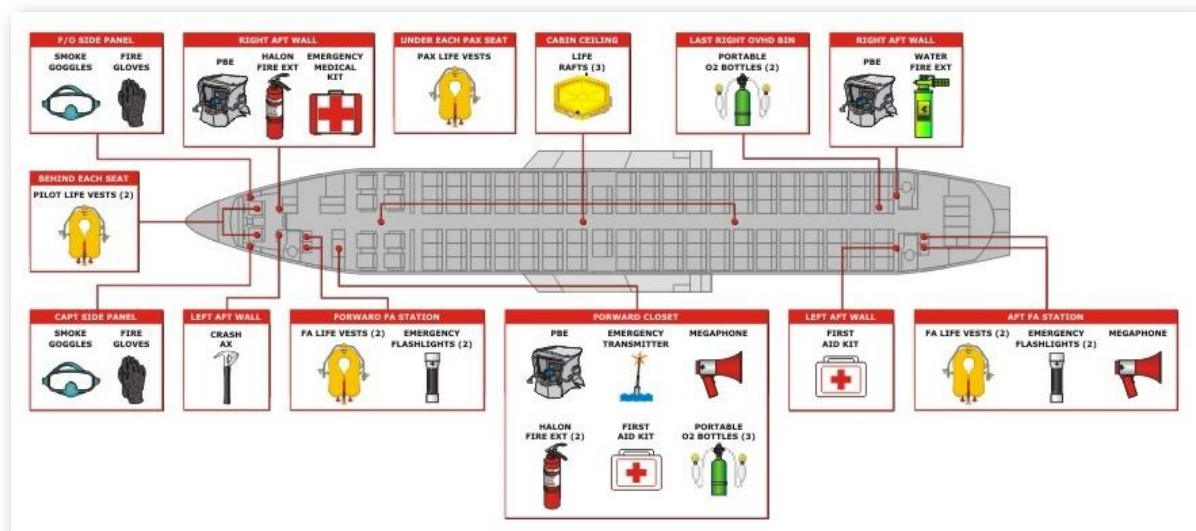


Figure 13: Overview of emergency equipment for a typical passenger aircraft

5.4 Rapid decompression

Since Hyperloop operates in a vacuum tube, the design has to account for a pressurization failure which, in the worst case, can lead to a rapid decompression of the cabin. From a structural point of view, this means that large pressure differentials can suddenly occur, which can create high loads on separation walls, the floor and other structures in the cabin. There is a risk that these structures fail and injure passengers or damage crucial systems. This topic shall receive adequate attention during the concept design of the Hyperloop.

6 Topics of High Importance

This chapter lists topics that are currently of the biggest concern to ADSE and that should receive a high priority.

6.1 Emergency Evacuation Scenario

Evacuating the Hyperloop vehicle in case of emergency is a topic that needs to be thoroughly addressed. The decisions can have a major impact on the design. The following aspects are of concern:

- Due to the tight fit of the vacuum tube around the Hyperloop vehicle, there seems to be no space to evacuate passengers through the six existing passengers doors. This is especially true when persons with reduced mobility are taken into account. As a consequence, evacuation through the forward and aft end of the pod seems to be the only feasible option, but it seems emergency exits are not foreseen there at this point. Since one of these evacuation routes can be blocked during an accident, for instance due to fire or a physical obstruction, the design shall enable both forward *and* aft evacuation.
- The evacuation takes place into a tube that is vacuum under normal operating conditions. The related safety scenarios shall be thoroughly addressed.

6.2 ECS Design and Pressurization

As the Hyperloop vehicle operates in a low pressure environment, the ECS system is more complex than that of a train. Two additional critical functions that the ECS needs to fulfil are:

1. Maintaining a safe and comfortable cabin pressure,
2. Maintaining a healthy climate regarding concentrations of oxygen and carbon dioxide.

These functions are also executed by the ECS of airplanes. However, these systems make use of so-called 'bleed air' from the engines, which will obviously not be available for the Hyperloop. Therefore, different systems must be used in order to accomplish these critical functions.

A drawback is that these new systems are probably not widely used and have a lower Technology Readiness Levels (TLR). This brings the need for thorough testing to show compliance to applicable safety requirements. Besides, since these functions are critical, it is important that single failures could not lead to failure of these functions. For example, this can be omitted by applying redundancy in the design.

Finally, since the passengers are traveling in this closed and pressurized environment, health aspects, like the amount of fresh air and the application of HEPA filters, need to receive adequate attention.

7 Conclusions and Recommendations

The Interior Evaluation by ADSE has resulted in the following conclusions:

- Since this document does not include an evaluation of the concept interior of Hyperloop, ADSE did not draw conclusions on the maturity of concept design. However, it is clear that HDP has created a good starting point for the top-level interior concept. HDP has a clear definition of the number of passengers per vehicle, the overall vehicle size and the intended level of service and comfort. There is also a concept floor plan. However, several important decisions are still to be made. Examples are the decision to integrate lavatories or not, and the concept for emergency evacuation. Both have a large impact on the Hyperloop interior and therefore need to receive adequate attention early in the project.
- Since the design of the Hyperloop interior is in the early stage of the concept phase, HDP has the opportunity to include best practices from Aerospace and Rail and make well-considered decisions on topics that have a high impact on the overall design. This reduces the risk of expensive changes at a late stage of the development process. This report can serve as important input for these decisions.

Recommendations:

- ADSE recommends to HDP to use the checklist of Chapter 4 as 'design guidelines' and use it to make sure all aspects of the design are addressed.
- ADSE recommends to focus on the emergency evacuation concept, as addressed in Paragraph 6.1. This has a high impact on the overall design of the interior lay-out.
- Although not an obvious part of the Hyperloop interior, the ECS system plays a critical role for the passengers. It also has an important interface with the interior since the air needs to be distributed throughout the cabin. It is important to give the ECS system adequate attention during the conceptual design phase, as explained in Paragraph 6.2.
- Finally, it's recommended that HDP looks at the interior from a perspective of 'Intended Use'. The transport of passengers is clearly the main use-case of the passenger version of Hyperloop, but there might be future scenarios that must be assessed already now. For instance:
 - Is combined transport of cargo and passengers foreseen, as this is quite common in aviation?
 - Is the transport of medical patients on stretchers considered? This might require a flexible approach to the seating arrangement and seat attachments.
 - Is it necessary to make the interior lay-out flexible enough to adapt to local (cultural) demands. For instance, in Asian countries, passengers don't like to travel backwards. Consequently, many trains have pivoting seats, and rail personnel rotates them at the terminal station before the train goes back in opposite direction.

Final note:

- The Hyperloop organization and the physical product are highly complex. In the fields of aerospace, rail and defense, it is common practice to use the methods of System Engineering (SE) for such projects. SE is a structured framework that helps to maintain a good overview of topics like requirements, risks and configuration management. A complex project like Hyperloop could benefit from such a structured approach. ADSE is happy to support HDP in this.

8 Next Phase and ADSE Contribution

8.1 Support overview

This report was written as part of 3-year support agreement between Hyperloop and ADSE, as presented to HDP in March 2021. This interior concept evaluation is the final deliverable of the first phase of the approach. The three phases of the approach can be summarized as follows:

Phase 1: Concept interior evaluation

- Evaluation of the current cabin interior concept.
- Provide a summary of points of attention for the next phase of the design.
- Deliverable: A report with a checklist of relevant interior topics and points of concern (this report).

Phase 2: Development of concept design and regulatory framework

- ADSE takes the lead in the design of the Hyperloop interior together with the consortium of partners.
- ADSE supports HDP in the development of regulatory framework, in cooperation with authorities.
- ADSE uses its experience from aerospace, tunnels and rail/metro to support the writing of a proposal for compliance demonstration.
- Deliverable: A feasible, producible and certifiable interior concept.
- Timeframe: Approximately 1 year.

Phase 3: Detailed design and compliance demonstration

- Detailed design of Hyperloop interior system, compliance demonstration, verification and validation.
- Creating a certification roadmap and supporting talks with regulators and notified bodies.
- Deliverable: A concept certification plan.
- Timeframe: Approximately 1 year.

8.2 Prerequisites for Phase 2

Before Phase 2 can start, the following prerequisites shall be in place:

- HDP shall freeze the top-level interior requirements and cabin lay-out. This includes important design drivers like the emergency evacuation philosophy and the decision to include a lavatory or not.

Annex 1 Input Documents

The following documents were shared by HPD with ADSE and were studied to execute this interior concept evaluation.

No.	Title	Date	Topic
01	HDP Partner Introduction meeting	?	Hyperloop development program & partner overview
02	HDP Partner Introduction meeting	?	Hyperloop development program & partner overview
03	HDP Workgroup Passenger	?	Passengerloop program overview
04	Controleprotocol Subsidieverlening	?	Subsidieverlening Rijksdienst voor Ondernemend NL
05	The Hyperloop Development Program	?	Deliverables, planning, milestones, interfaces
06	HDP Partner Introduction meeting	26-02-21	Deliverables, planning, milestones, interfaces
07	HDP Planning [Excel doc.]	?	Planning of HDP deliverables
08	Acceptance of Hyperloop	Nov 2020	Hyperloop acceptance among potential users
09	Airlock	?	Stations (boarding/unboarding)
10	HDP Phase 2, Program Plan	26-11-20	Planning
11	Hardt in cooperation with HFT Stuttgart	21-04-21	Stakeholder Needs
12	HDP Phase 2, Governance Plan	24-11-20	Schedule
13	Hardt Hyperloop Transport Hub	07-09-18	Hyperloop Station
14	Reporting Requirements for HDP	Oct 2020	Finance and funds
15	Hyperloop Interior	?	Requirements & concept design
16	Hyperloop parameters [Excel doc.]	?	Operational Control Concept
17	HARDT station layout	?	Stations (boarding/unboarding)
18	Cost Study: Comparison Hyperloop - HSR	07-02-19	BAM: Cost comparison HS Rail
19	Pre-feasibility Study Schiphol-Hyperloop	07-02-20	commercial and economic viability
20	Pre-feasibility Study Schiphol-Hyperloop	07-02-20	Policy, strategic stakeholder analysis & prioritization
21	Passenger journey	?	Service design
22	Roland Berger for Kingdom Saudi Arabia	24-03-21	Broad system overview
23	Service Design Sprint Report	?	Stakeholder Needs
24	The Hyperloop	13-09-19	Hyperloop Vision document
25	Prospective LCA of Hyperloop system	Feb 2020	Lifecycle environmental impact of Hyperloop